



QUALITY IN SPORT

eISSN 2450-3118 · Open Access · Peer-reviewed

apcz.umk.pl/QS Nicolaus Copernicus University in Toruń



LIS, Krzysztof, GAŁWIACZEK, Jakub, DUDEK, Jarosław, GREBOSZ, Damian, LYSAK, Aleksandra Kinga, KALISZ, Aleksandra and ILKIEWICZ, Lucja. Effects of Fixture Congestion on Post-Match Recovery Kinetics in Soccer Players: A Narrative Review. *Quality in Sport*. 2026;72:75609. <https://doi.org/10.12775/QS.2026.72.75609>

ARTICLE TIMELINE

Received: 23.08.2026. Revised: 30.08.2026. Accepted: 07.09.2026. Published: 16.09.2026.

The journal has been awarded 20 points in the parametric evaluation by the Polish Ministry of Higher Education and Science (Annex to the announcement of 05.01.2024, No. 32553). Unique Journal Identifier: 201398. Scientific disciplines assigned: Economics and Finance (Field of Social Sciences); Management and Quality Sciences (Field of Social Sciences).

Punkty Ministerialne z 2019 – aktualny rok 20 punktów. Załącznik do komunikatu Ministra Szkolnictwa Wyższego i Nauki z dnia 05.01.2024 Lp. 32553. Posiada Unikatowy Identyfikator Czasopisma: 201398. Przypisane dyscypliny naukowe: Ekonomia i finanse (Dziedzina nauk społecznych); Nauki o zarządzaniu i jakości (Dziedzina nauk społecznych). © The Authors 2026.

OPEN ACCESS · CC BY-NC-SA 4.0 This article is published with open access under the License Open Journal Systems of Nicolaus Copernicus University in Toruń, Poland, and is distributed under the terms of the Creative Commons Attribution Non-commercial Share Alike License (<http://creativecommons.org/licenses/by-nc-sa/4.0/>), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the work is properly cited. The authors declare no conflict of interest regarding the publication of this paper.

Effects of Fixture Congestion on Post-Match Recovery Kinetics in Soccer Players: A Narrative Review

Krzysztof Lis, ORCID: <https://orcid.org/0009-0008-0803-1852>

E-mail: lis.krzysztof.123@gmail.com

Faculty of Medicine, Jan Kochanowski University in Kielce, al. IX Wieków Kielc 19a, 25-516
Kielce, Poland

Jakub Gałwiazek, ORCID: <https://orcid.org/0009-0005-0912-2594>

E-mail: oggalwiazek@gmail.com

Provincial Integrated Hospital, ul. Grunwaldzka 45, 25-736 Kielce, Poland

Jarosław Dudek, ORCID: <https://orcid.org/0009-0005-1273-5682>

E-mail: jaro803dud@gmail.com

St. Raphael's Voivodeship Specialist Hospital in Czerwona Góra, ul. Czerwona Góra 10, 26-060
Chęciny, Poland

Damian Grębosz, ORCID: <https://orcid.org/0009-0000-3046-4701>

E-mail: damian.grebosz1999@gmail.com

Faculty of Medicine, Jan Kochanowski University in Kielce, al. IX Wieków Kielc 19a, 25-516
Kielce, Poland

Aleksandra Kinga Łysak, ORCID: <https://orcid.org/0009-0001-7566-4368>

E-mail: aleksandra.lysak@wp.pl

Provincial Integrated Hospital, ul. Grunwaldzka 45, 25-736 Kielce, Poland

Aleksandra Kalisz, ORCID: <https://orcid.org/0009-0004-5384-7309>

E-mail: akalisz@interia.eu

Faculty of Medicine, Jan Kochanowski University in Kielce, al. IX Wieków Kielc 19a, 25-516
Kielce, Poland

Łucja Ilkiewicz, ORCID: <https://orcid.org/0009-0000-3903-9920>

E-mail: lucja571@wp.pl

Faculty of Medicine, Jan Kochanowski University in Kielce, al. IX Wieków Kielc 19a, 25-516
Kielce, Poland

Corresponding Author

Krzysztof Lis, E-mail: lis.krzysztof.123@gmail.com

Abstract

Background. Fixture congestion is common in soccer and reduces the time available for recovery between matches. Recovery is multidimensional, and physical, neuromuscular, biochemical, and perceptual markers may normalize at different rates.

Aim. This narrative review aimed to synthesize evidence on the effects of fixture congestion on post-match recovery kinetics in soccer players.

Material and methods. A predefined full-text literature set relevant to fixture congestion, fatigue, post-match recovery, workload, performance, and injury risk was synthesized qualitatively. Original studies formed the primary evidence base, with review articles used for context.

Results. Recovery was heterogeneous. Several neuromuscular, muscular, biochemical, and perceptual variables remained altered at 48 h, while selected performance measures recovered by 72 h. Congested schedules and high match exposure were associated with slower recovery of selected outcomes even when performance was preserved.

Conclusions. A return of physical performance does not necessarily indicate complete recovery. Fixed 48- or 72-h intervals should not be interpreted alone as evidence of full restoration. Monitoring should integrate individual match exposure, workload, performance, and perceptual responses.

Key words: fixture congestion; soccer; football; post-match recovery; fatigue; neuromuscular fatigue; recovery kinetics; training load.

1. Introduction

The competitive demands of modern soccer frequently require players to participate in several matches within a relatively short period. This situation is particularly common among professional teams competing simultaneously in domestic leagues, national cup competitions and international tournaments. Fixture congestion has been defined as a sequence of at least two successive matches separated by less than 96 hours of recovery [1]. In practice, this often results in teams playing two or even three matches within seven days, substantially reducing the time available for training and recovery between competitive events [2].

The increasing frequency of competitive matches has made fixture congestion an important issue in professional soccer. Elite players may be exposed to demanding periods in which consecutive matches are separated by only two or three recovery days. Lundberg and Weckström reported that top-level players may complete approximately 50–70 competitive matches during a season, while the recovery period before the next match may sometimes be limited to approximately 72 hours [3]. The challenge created by this scheduling pattern is not restricted to professional senior players. Studies included in the present review have also examined subelite and youth populations exposed to multiple matches within the same week [4–5].

An increased number of matches also modifies the structure of the weekly training microcycle. Oliveira et al. demonstrated differences in internal and external training loads between one-, two- and three-match weeks in elite soccer players [2]. During congested periods, coaching staff must therefore balance the need to prepare players for the next match against the need to limit additional training stress and facilitate recovery. Similarly, match participation itself contributes substantially to the individual load accumulated by players, meaning that starters and players with greater match exposure may experience a different recovery challenge than players who participate for fewer minutes or do not play [6].

Although congested schedules intuitively appear likely to impair subsequent performance, findings are not entirely consistent. A systematic review by Julian et al. found that total distance covered during professional match-play was not significantly affected by fixture congestion, while effects on other physical-performance variables were equivocal [1]. Maintenance of match-running performance, however, does not necessarily demonstrate complete physiological recovery. This distinction is particularly important because players may retain their ability to perform during subsequent competition despite the persistence of fatigue or other physiological disturbances.

A soccer match imposes substantial physical and physiological demands and can produce fatigue that persists beyond the immediate post-match period. Recovery following match-play is therefore not a single process, but involves the restoration of neuromuscular function, physical performance, physiological and biochemical status, and perceived readiness. Importantly, these components do not necessarily recover at the same rate.

Nédélec et al. examined professional players before a competitive match and at 24, 48 and 72 hours afterward. Reductions in countermovement jump performance, hamstring strength and peak sprint speed, together with increased muscle soreness, were observed during the recovery period. The magnitude of some recovery disturbances was also associated with specific match actions, including short sprints and hard changes of direction [7]. These findings indicate that the time course of recovery may depend not only on the time elapsed after a match but also on the demands experienced by an individual player during competition.

Neuromuscular fatigue may remain evident for several days after match-play. Thomas et al. investigated semi-professional players following simulated soccer match-play and reported that maximal voluntary contraction remained impaired at 72 hours. Peripheral fatigue had also not fully recovered at this time point, while central fatigue remained detectable at 48 hours [8]. Competitive

match-play has produced similarly complex recovery patterns. Brownstein et al. observed that jump performance recovered by 48 hours, whereas subjective fatigue and reduced motivation to train could persist at 72 hours even when neuromuscular and physical-performance measures had returned toward baseline [9].

The discrepancy between different recovery markers is also evident when biochemical and performance responses are considered together. Marqués-Jiménez et al. assessed players after a 72-hour recovery period and found physical-performance measures similar to pre-match values, while biochemical indicators continued to demonstrate signs of physiological stress. In particular, cortisol remained elevated, and the testosterone-to-cortisol ratio remained reduced compared with baseline [10]. Thus, the restoration of sprinting, jumping or other physical capabilities cannot automatically be interpreted as complete recovery of all physiological systems.

These findings underline the multidimensional character of post-match recovery. The use of a single performance or physiological variable may provide only a partial representation of a player's recovery status. Consequently, the interpretation of recovery should consider the potentially different time courses of physical, neuromuscular, biochemical and perceptual responses.

The time course of post-match recovery becomes particularly important when another competitive match is scheduled before all recovery processes have been completed. Under regular conditions, residual fatigue may gradually resolve over several days. During fixture congestion, however, the next match can occur within the same period in which neuromuscular impairment, muscle soreness or physiological stress is still present.

Lundberg and Weckström directly compared recovery following a congested three-match week with recovery following a regular one-match week in professional players. Although sprint and jump performance were not clearly impaired by the congested cycle, greater muscle soreness and increased plasma urea responses indicated that some signs of physiological stress remained after the congested period [3]. This finding again illustrates that preserved physical performance does not necessarily indicate that recovery is complete.

Repeated exposure may also produce cumulative effects that are not immediately apparent after a short congested period. Rollo et al. compared players completing one or two competitive matches per week over six weeks. No clear impairment in physical-test performance was identified after three weeks, whereas after six weeks the group playing two matches per week demonstrated poorer countermovement jump, sprint and repeated high-intensity exercise performance [4]. These

observations suggest that the consequences of fixture congestion may depend not only on the interval between individual matches but also on the duration of repeated exposure to a congested schedule.

Recent evidence further supports the importance of distinguishing recovery after regular and congested match weeks. Wang et al. compared post-match responses between congested and non-congested schedules and found lower countermovement jump performance as well as greater pain sensitivity and soreness during the 24- and 48-hour recovery period in the congested condition [11]. The authors concluded that fixture congestion was associated with delayed neuromuscular and sensory recovery.

Incomplete recovery may also be relevant to player health. Dupont et al. observed a higher injury rate when professional players participated in two matches per week compared with one match per week, despite no significant reduction in the principal match-running performance variables [12]. Research focusing specifically on injury during fixture congestion has subsequently shown that injury risk tends to be higher during congested match periods, although findings differ according to study design, competition characteristics and the definition of congestion used [13]. The relationship between incomplete recovery and injury should therefore be interpreted cautiously, but it reinforces the importance of understanding the recovery status of players exposed to short inter-match intervals.

Taken together, the available evidence indicates that fixture congestion cannot be evaluated solely through subsequent match performance. Players may preserve selected measures of external performance while other indicators of recovery remain altered. The recovery interval itself, cumulative match exposure, individual playing demands and training performed between matches may all contribute to the recovery pattern. Examining recovery kinetics therefore provides a broader perspective on the physiological consequences of congested competition than assessment of match performance alone.

The aim of this narrative review is to synthesize the available evidence on the effects of fixture congestion on post-match recovery kinetics in soccer players, with particular emphasis on neuromuscular, physical, biochemical, perceptual and workload-related responses. The review also aims to examine whether commonly occurring inter-match recovery periods of approximately 48–72 hours are sufficient for the restoration of players' pre-match status.

A secondary aim is to consider the potential implications of incomplete recovery for subsequent physical performance, training-load management and injury risk. Particular attention is given to differences in the time course of individual recovery markers, since restoration of physical performance may occur before complete normalization of other physiological or perceptual indicators.

2. Research materials and methods

This study was conducted as a narrative review focusing on the effects of fixture congestion on post-match recovery kinetics in soccer players. The literature considered for the review was restricted to a predefined full-text source set relevant to fixture congestion, match-related fatigue, post-match recovery, training and match load, subsequent performance, and injury risk. Original investigations constituted the primary evidence used to describe recovery responses, while review articles were used to provide broader context and to compare findings reported across individual studies. Because the objective was to describe and interpret the time course of recovery across different physiological and performance domains rather than to calculate pooled effect estimates, the available evidence was synthesized narratively.

The included literature used different approaches to define and investigate fixture congestion. In professional soccer, fixture congestion has been operationalized as at least two successive matches separated by less than 96 h [1]. Other studies compared one-, two-, and three-match weekly schedules [2] or directly contrasted congested and regular competitive weeks [11]. These different study designs were retained because they reflected the range of competitive schedules represented in the available literature.

The original database search procedures used to assemble the predefined literature set, including the databases searched, exact search dates, and complete search strings, were not documented as part of the present review process. These details are therefore not reconstructed or reported. No publications outside the predefined source set were introduced into the synthesis.

The available publications were evaluated for their relevance to the main objective of the review: to examine how fixture congestion influences the time course of post-match recovery in soccer players. Particular attention was given to studies investigating multiple matches within a short period, differences between congested and non-congested schedules, different numbers of matches within a weekly microcycle, and recovery responses assessed during the hours and days following match-play.

The evidence base included studies using repeated post-match measurements at 24, 48, and 72 h [7–9], investigations comparing different competitive schedules [2,11], and studies examining recovery after approximately 72 h [10]. These designs allowed the recovery process to be considered across different time points and competitive contexts.

Publications were considered relevant to the narrative synthesis when they involved soccer players and addressed at least one aspect directly related to fixture congestion, match-related fatigue, or post-match recovery. Studies involving professional, semiprofessional, subelite, and youth soccer players were retained because the predefined literature set included populations from different competitive levels.

Original studies were included in the synthesis when they examined exposure to multiple matches within a short period, differences between congested and non-congested schedules, recovery responses following soccer match-play, internal or external workload associated with different competitive schedules, or physical, neuromuscular, biochemical, physiological, or perceptual responses during the post-match recovery period.

Particular attention was given to studies assessing recovery during the first 24–72 h after match-play, because this interval represents a central component of the present review. Nédélec et al. assessed physical, biochemical, and perceptual responses at 24, 48, and 72 h after competitive match-play [7]. Thomas et al. [8] and Brownstein et al. [9] used repeated measurements to characterize neuromuscular and physical recovery, while Marqués-Jiménez et al. examined physical and biochemical responses after a 72-h recovery period [10].

Relevant outcomes included countermovement jump performance, sprint and repeated-sprint performance, maximal voluntary contraction, muscle strength, creatine kinase, urea, inflammatory and hormonal responses, perceived fatigue, muscle soreness, pain-related measures, and perceived recovery. Studies assessing match exposure, training load, subsequent match performance, or injury occurrence were also considered when these outcomes contributed to the interpretation of recovery during congested schedules.

Review articles were retained when they directly addressed an aspect of the research question and provided context for interpreting findings from original investigations. For example, existing reviews have examined the effects of fixture congestion on match performance [1] and injury occurrence [13]. These publications were used as supporting evidence and were not treated as independent primary observations.

Publications unrelated to soccer match congestion, post-match recovery, fatigue, workload, performance, or injury were not incorporated into the narrative synthesis. Studies involving sports other than soccer were not included in the evidence base.

Information relevant to the objectives of the review was extracted from the included publications and organized according to the characteristics of the study and the recovery variables investigated. For original studies, attention was given to participant characteristics and competitive level, competitive schedule or match exposure, the interval between matches, post-match assessment time points, outcome measures, and findings relevant to recovery.

Because the principal focus of the review was recovery kinetics, particular attention was paid to the timing of post-match assessments. Findings were therefore considered in relation to the time elapsed after competition, with emphasis on the immediate post-match period and on responses observed at approximately 24, 48, and 72 h. This approach allowed studies using repeated measurements to be compared with investigations assessing recovery at a single clinically or practically relevant time point [7–11].

The extracted findings were organized into thematic domains corresponding to the objectives of the review: neuromuscular and physical recovery, biochemical and physiological responses, perceptual recovery, and the influence of match exposure and training load. Subsequent match performance and injury-related outcomes were considered separately when they contributed to understanding the potential consequences of incomplete recovery.

The findings were compared qualitatively across studies, with attention to both consistent and conflicting observations. Particular consideration was given to differences in the recovery time course between individual outcome measures. This was important because the included literature indicates that restoration of physical performance may not always coincide with normalization of physiological or perceptual markers [9–10]. Similarly, direct comparisons between congested and regular schedules indicate that fixture density may modify the magnitude and time course of post-match recovery [11].

No quantitative pooling of results was performed. The available evidence was instead integrated through narrative synthesis to describe the pattern and timing of recovery across the different outcome domains and to examine whether the commonly encountered 48–72 h interval between matches appears sufficient for recovery of the variables investigated.

3. Research results

3.1. Characteristics of Fixture Congestion and Recovery Intervals

The studies included in this review used different approaches to define and investigate fixture congestion. The systematic review by Julian et al. considered fixture congestion as a sequence of at least two successive matches separated by less than 96 h [1]. Individual studies applied several competitive models, including one versus two matches per week [4,12], one versus three matches within a week [5], two or more matches within a seven-day period [6], and three-match weekly microcycles [3,14].

Rollo et al. followed subelite players over six weeks and compared schedules involving either one or two competitive matches per week [4]. Physical and subjective assessments were performed 48 h after match-play at baseline and after three and six weeks. No between-group differences in physical performance were found after three weeks, whereas after six weeks the players completing two matches per week showed lower CMJ and Yo-Yo Intermittent Recovery Test performance and slower 10- and 20-m sprint times [4].

Oliveira et al. examined training load during one-, two-, and three-match weekly microcycles in elite professional players [2]. Hernández-Davó et al. used a different design, assessing youth players 72 h after either a one-match or a three-match period. In the three-match period, the matches were separated by 72 and 96 h, and several physical-performance variables were lower than after the one-match period [5].

Shorter recovery intervals were examined in both simulated and competitive settings. Page et al. used three soccer-specific match simulations performed over five days, with 48 h separating each trial [15]. Mohr et al. studied a three-game microcycle in which 72 h separated the first and second games and 96 h separated the second and third games [14]. Several recovery variables showed a different response after the second match, which followed the shorter inter-match interval.

Lundberg and Weckström assessed professional players 72 h after the final match of a three-match week and compared these responses with measurements obtained 72 h after a match during a regular one-match week [3]. Players with high exposure during the congested cycle showed greater changes in muscle soreness and plasma urea, while no effects were observed for sprint or jump performance.

Longer congested periods were investigated by Dellal et al., who examined three periods consisting of six consecutive matches separated by approximately three days [16]. Physical and technical match-performance measures did not differ significantly across the successive matches.

Tournament competition provided another model of fixture congestion. Wilke et al. analyzed the Brazilian Under-20 national team during a tournament in which the first phase consisted of four matches separated by 48 h and the second phase consisted of five matches separated by 72 h [17]. Match load and perceptual outcomes were broadly similar between the two phases, while post-match biochemical responses differed according to the recovery interval. Wang et al. compared a congested week involving two matches within seven days and separated by no more than three days with a regular week containing one match and at least six days between matches [11].

The available studies therefore covered several forms of fixture congestion, ranging from 48-h inter-match intervals to repeated two- or three-match weeks and longer periods of successive competition.

3.2. Neuromuscular and Physical Recovery

3.2.1. Jump Performance

Countermovement jump performance was frequently used to assess post-match neuromuscular recovery. Nédélec et al. measured CMJ before a competitive match and at 24, 48, and 72 h afterward. CMJ performance decreased during the recovery period, and the reduction observed at 24 h was significantly related to the number of hard changes of direction completed during the match [7].

Thomas et al. also observed a prolonged recovery pattern in jump performance following simulated match-play [8]. In competitive match-play, Brownstein et al. reported that jump performance required approximately 48 h to recover [9]. Their neuromuscular measurements showed that the recovery of central and peripheral function also occurred over the first 48 h after competition.

A different pattern emerged when match exposure was increased. Hernández-Davó et al. found lower CMJ performance 72 h after a three-match period than after a one-match period in young players [5]. Mean CMJ height was 39.94 ± 4.49 cm after the one-match period and 36.05 ± 4.83 cm after the three-match period [5].

Wang et al. reported differences between congested and regular weeks during the short-term recovery period [11]. Pre-match and immediate post-match CMJ values did not differ significantly

between week types, but CMJ height was significantly lower during the congested week at 24 h and 48 h after the match [11].

Not all congested-schedule studies found persistent impairment. Lundberg and Weckström observed no effect of the three-match cycle on jump performance when players were assessed 72 h after the final match [3]. Marqués-Jiménez et al. similarly found no significant difference between baseline and 72-h post-match squat-jump performance after a single match [10].

Franceschi et al. reported another recovery pattern in Italian Serie A youth players [18]. CMJ measures were impaired immediately after match-play but had returned to baseline by 48 h. At the same time point, isometric posterior-chain performance and muscle soreness remained impaired [18].

3.2.2. Sprint and Repeated-Sprint Performance

Sprint-related recovery differed across studies and according to the type of assessment. Nédélec et al. reported a significant reduction in peak sprint speed during the 72-h recovery period following competitive match-play [7].

Rollo et al. observed no difference in sprint performance between one- and two-match-per-week groups after three weeks. After six weeks, however, the two-match group recorded slower 10-m and 20-m sprint times and lower Yo-Yo Intermittent Recovery Test performance [4]. The corresponding between-group differences at six weeks were significant for CMJ, the Yo-Yo test, and both sprint distances.

Hernández-Davó et al. reported slower sprint split times after the three-match period [5]. Although the overall 25-m sprint result was less clear, performance was impaired over the 10–15 m, 15–20 m, and 20–25 m sections, with the largest effect observed over the final 5-m split.

Repeated-sprint performance was also affected in the three-game microcycle investigated by Mohr et al. [14]. Performance remained impaired during the first two recovery days following the first match and was still reduced on the third recovery day following the second and third matches. The impairment was more pronounced during parts of the recovery period following the second game than after the first [14].

In contrast, Marqués-Jiménez et al. found no significant differences between baseline and 72-h post-match values for total repeated-sprint time or the repeated-sprint fatigue index [10].

3.2.3. Muscle Strength and Contractile Function

Muscle strength and contractile function showed prolonged changes in several studies. Nédélec et al. reported reduced isometric hamstring strength during the 72-h post-match recovery period [7]. Thomas et al. identified both central and peripheral components of neuromuscular fatigue after simulated match-play [8]. Maximal voluntary contraction remained below baseline at 72 h. Voluntary activation was reduced immediately after exercise and remained depressed at 48 h, while quadriceps potentiated twitch force was reduced after the simulation and had not fully recovered at 72 h [8].

The recovery time was shorter in the competitive match studied by Brownstein et al. [9]. Maximal voluntary contraction was reduced immediately after match-play and remained lower at 48 h, but had recovered by 72 h. Voluntary activation and potentiated twitch force returned to baseline by 48 h [9].

Page et al. reported changes in knee-flexor function during three match simulations separated by 48 h [15]. Peak torque at $300^{\circ}\cdot\text{s}^{-1}$ was lower in the second and third simulations than in the first, while peak torque at $60^{\circ}\cdot\text{s}^{-1}$ was lower in the third trial. Muscle soreness was also higher during the third trial [15].

Franceschi et al. observed incomplete recovery of isometric posterior-chain performance at 48 h after match-play in youth players, even though CMJ and perceived fatigue had returned to baseline [18].

3.3. Biochemical and Physiological Recovery

3.3.1. Markers of Muscle Damage

Creatine kinase was included as a marker of muscle damage in several of the studies. Nédélec et al. monitored CK together with neuromuscular, physical, and perceptual measures across the 72-h recovery period [7].

Oliveira et al. included CK measurements alongside internal and external training-load monitoring during one-, two-, and three-match weekly microcycles [2]. Differences in CK were not consistent across all analyzed microcycles; a significant difference was reported between two of the individual microcycles included in the study [2].

Mohr et al. found marked elevations in indicators of muscle damage during their three-match microcycle [14]. CK responses were greater during parts of the recovery period following the

second game than following the first or third games. Muscle soreness and changes in knee-joint range of motion showed a similar pattern [14].

Lundberg and Weckström found a greater increase in plasma urea among players with high match exposure following the congested three-match week than among control players [3]. Sprint and jump performance did not differ, while changes in several other blood variables were trivial, unclear, or absent [3].

Wilke et al. found higher CK responses during the tournament phase in which matches were separated by 48 h than during the phase with 72-h intervals [17]. Importantly, blood samples were collected approximately 13–15 h after each match; the 48- and 72-h values referred to the interval between matches rather than the timing of blood collection [17].

3.3.2. Inflammatory and Immune Responses

Mohr et al. examined inflammatory and immune responses throughout the three-match microcycle [14]. Match-play produced changes in several inflammatory and immune-related variables, and responses to muscle damage and inflammation were greater during parts of the recovery period following the second game. This game followed a 72-h interval, compared with 96 h before the third game [14].

Wilke et al. also found differences in inflammatory responses according to fixture spacing [17]. The change in C-reactive protein was greater after several matches from the tournament phase characterized by 48-h intervals than after matches from the 72-h phase [17]. External match-load variables and perceptual measures did not show corresponding differences in the same comparison. Lundberg and Weckström did not find evidence that immune function was compromised 72 h after the three-match cycle, although muscle soreness and selected markers of physiological stress differed between highly exposed players and controls [3].

3.3.3. Hormonal Responses

Hormonal responses were examined less frequently than physical or neuromuscular variables. Mohr et al. assessed cortisol and testosterone during their repeated-match protocol [14]. Cortisol increased immediately after the first match and returned to baseline after one recovery day. Following the second game, cortisol was also elevated immediately after match-play and remained elevated one day into recovery [14].

Marqués-Jiménez et al. assessed testosterone, cortisol, the testosterone-to-cortisol ratio, and urea before a match and again after 72 h [10]. Testosterone and urea were not significantly different

from baseline after the recovery period. Cortisol was significantly higher, while the testosterone-to-cortisol ratio was significantly lower. At the same time, squat-jump and repeated-sprint performance did not differ significantly from baseline [10].

3.4. Perceptual Recovery

3.4.1. Fatigue and Perceived Recovery

Subjective recovery measures did not always follow the same pattern as physical-performance variables. Rollo et al. reported no significant changes over time or time-by-group interactions for stress, recovery, or the Stress Recovery Index, despite significant deterioration in several objective performance measures after six weeks of two matches per week [4].

Brownstein et al. found that perceived fatigue persisted at 72 h after competitive match-play, whereas the measured central and peripheral neuromuscular disturbances had recovered by 48 h and maximal voluntary contraction had recovered by 72 h [9].

In the youth players studied by Franceschi et al., perceived fatigue had returned to baseline at 48 h, while posterior-chain muscle function and muscle soreness remained impaired [18].

Wilke et al. did not observe significant differences in pre-match recovery or soreness when the same players were compared between matches occurring within the 48- and 72-h tournament schedules [17].

3.4.2. Muscle Soreness and Pain-Related Measures

Muscle soreness showed prolonged post-match changes in several investigations. Nédélec et al. reported increased soreness during the 72-h recovery period. The increase was significantly related to the number of short sprints performed during the match at both 48 and 72 h [7].

Page et al. found higher muscle soreness during the third of three soccer-specific simulations separated by 48 h compared with the first simulation [15]. Mohr et al. also observed marked changes in delayed-onset muscle soreness during repeated match-play, with the longest-lasting responses occurring after the second game in several measurements [14].

Franceschi et al. found that muscle soreness remained above baseline at 48 h after match-play [18]. Wang et al. extended perceptual assessment by combining visual analogue scale scores with pressure pain sensitivity and CMJ performance [11]. In the congested week, CMJ performance was lower at 24 and 48 h, while pain-related and perceptual measures also differed from the regular week during recovery [11].

3.5. Effect of Recovery Interval Duration

3.5.1. Recovery Within 24 Hours

The first 24 h after match-play were associated with disturbances in several recovery variables. Nédélec et al. found reductions in CMJ and hamstring strength during this period, and the CMJ reduction at 24 h was associated with the number of hard changes of direction performed during the match [7].

Thomas et al. observed substantial central and peripheral neuromuscular fatigue immediately after simulated match-play. Potentiated twitch force remained markedly reduced at 24 h [8]. Brownstein et al. also reported reductions in voluntary activation and muscle contractile function at 24 h following competitive match-play [9].

During congested competition, Wang et al. found significantly lower CMJ performance at 24 h in the congested week than in the regular week [11]. Pain-related and subjective responses also remained more pronounced during this phase of recovery [11].

Mohr et al. found substantial changes in muscle soreness, muscle-damage markers, inflammatory responses, and repeated-sprint ability during the early post-match recovery period across their three-game protocol [14].

3.5.2. Recovery After 48 Hours

By 48 h, recovery differed according to the outcome measured. Brownstein et al. found that voluntary activation, potentiated twitch force, and jump performance had recovered by this time, although maximal voluntary contraction remained slightly reduced and perceived fatigue persisted for longer [9].

In the simulated match protocol used by Thomas et al., central fatigue remained detectable at 48 h, while peripheral contractile function followed a slower recovery course and had not fully recovered by 72 h [8].

Franceschi et al. found a mixed recovery profile at 48 h [18]. CMJ and perceived fatigue had returned to baseline, whereas isometric posterior-chain performance and muscle soreness had not. At this point, players were allocated to either a complete or reduced training session, resulting in clearly different training loads between groups on MD+2 [18].

Wang et al. reported significantly lower CMJ height during the congested week at 48 h compared with the regular week [11]. Pressure pain sensitivity and VAS responses also differed between the two schedules during the recovery period.

Wilke et al. did not measure recovery markers exactly 48 h after an individual match. Instead, they compared matches occurring in tournament phases with either 48- or 72-h inter-match intervals. Blood collected 13–15 h after the matches showed greater CK and C-reactive protein responses during the 48-h phase, despite similar external match loads and perceptual responses [17].

3.5.3. Recovery After 72 Hours

Results at 72 h differed substantially between studies. Nédélec et al. recorded alterations in CMJ, isometric hamstring strength, peak sprint speed, and muscle soreness during the 72-h recovery period [7]. Thomas et al. found that maximal voluntary contraction and peripheral contractile function had not fully recovered at 72 h following simulated match-play [8].

Brownstein et al. reported a faster recovery after competitive match-play. Maximal voluntary contraction had returned to baseline by 72 h, and the measured central and peripheral components of neuromuscular fatigue had recovered earlier, although perceived fatigue remained elevated at 72 h [9].

Following a three-match period, Hernández-Davó et al. found lower CMJ performance, impaired change-of-direction performance, reduced ankle dorsiflexion range of motion, and slower sprint split times at 72 h compared with the one-match period [5].

Lundberg and Weckström found no difference in sprint or jump performance 72 h after a congested three-match week compared with a regular one-match week. Muscle soreness and plasma urea, however, showed greater changes in the players with high match exposure [3].

Marqués-Jiménez et al. also found physical-performance values comparable with baseline after 72 h, while cortisol remained elevated and the testosterone-to-cortisol ratio remained reduced [10].

3.6. Match Exposure and Training Load as Modifiers of Recovery

Match exposure and the work performed during competition were associated with subsequent recovery in several studies. Nédélec et al. found that the number of short sprints was related to muscle soreness at 48 and 72 h, while hard changes of direction were associated with the reduction in CMJ performance at 24 h [7].

Garcia et al. monitored 29 professional players over 119 training sessions and 33 matches [6]. Weeks were classified as congested when they contained at least two matches and as regular when they contained one match or fewer. Players were also grouped according to match participation: at least 60 min, less than 60 min, or no participation. Regular weeks showed higher values for several

acute, monotony, and strain indices than congested weeks, while players completing at least 60 min generally accumulated the highest workload values [6].

Oliveira et al. also reported differences in the distribution of internal and external training load across one-, two-, and three-match microcycles [2]. Overall weekly sRPE did not differ significantly between the one-, two-, and three-match schedules, despite differences in the distribution of load across individual training days [2].

The training performed during recovery was examined directly by Franceschi et al. [18]. At 48 h post-match, players had not fully recovered posterior-chain strength or muscle soreness. The complete-training group subsequently completed a greater external and internal load than the reduced-training group on MD+2 [18].

Howle et al. compared single- and multi-match weeks in professional players [19]. External training-load measures were lower during multi-match weeks, and internal and external training loads were also lower during seasons containing more congested scheduling. Total weekly loads remained comparable between single- and multi-match weeks because match exposure contributed a greater proportion of the total load [19].

Wilke et al. found no significant differences in several external match-load variables between the 48- and 72-h tournament phases, while biochemical responses differed between the two schedules [17].

3.7. Fixture Congestion and Subsequent Match Performance

Several studies reported maintenance of match-running performance during fixture congestion. Dupont et al. found no significant differences in total distance, high-intensity distance, sprint distance, or number of sprints between periods in which professional players completed one or two matches per week [12]. Injury incidence, however, was higher during the two-match schedule [12]. Dellal et al. also found no significant changes in physical or technical match performance across six successive matches during congested periods [16]. Total distance and distances covered at different intensities remained similar across matches, as did successful passes, balls lost, touches per possession, and duels won [16].

The systematic review by Julian et al. found no significant overall effect of fixture congestion on total distance covered during professional male match-play, while findings for several other running-performance variables were inconsistent across the available studies [1].

Other investigations identified changes outside standard match-running measures. Mohr et al. reported lower high-intensity running during the second game of their three-match protocol and slower recovery of repeated-sprint performance following this match [14]. Rollo et al. observed impairments in controlled measures of jumping, sprinting, and repeated high-intensity exercise after six weeks of playing two matches per week [4], despite the absence of such differences after three weeks.

Lundberg and Weckström reported no reduction in sprint or jump performance 72 h after a congested three-match week, although muscle soreness and plasma urea responses differed in highly exposed players [3].

Thus, the studies included in the review reported instances in which external match-running performance remained stable while other physical, perceptual, biochemical, or injury-related outcomes differed between schedules.

3.8. Incomplete Recovery and Injury Risk

Several studies examined injury occurrence during periods of fixture congestion. Dupont et al. reported an injury incidence of 25.6 injuries per 1000 h of exposure when players completed two matches per week compared with 4.1 injuries per 1000 h when they completed one match per week. The running-performance variables analyzed in the study did not differ significantly between the two schedules [12].

Howle et al. also reported higher injury incidence during multi-match weeks [19]. Total injury rates were greater during multi-match than single-match weeks, and injury incidence was highest in the seasons containing the greatest amount of congested scheduling. At the same time, external training loads were lower during multi-match weeks [19].

Bengtsson et al. analyzed data from 27 professional European teams over 11 seasons [20]. The dataset included 8150 matches and more than one million hours of football exposure. In league matches, total and muscle injury rates were higher when matches were preceded by four or fewer recovery days than when at least six recovery days were available. Hamstring and quadriceps injury rates were also higher with the shorter recovery interval. Increased match load was associated with a higher muscle-injury rate within the same match sequence [20].

Dellal et al. reported a different pattern when training and match exposure were considered separately [16]. Overall injury incidence did not differ significantly between congested and non-congested periods. Match injury incidence was higher during congested periods, whereas training

injury incidence was lower. Match injury rates were 43.3 and 18.6 injuries per 1000 h of exposure during congested and non-congested periods, respectively [16].

Carling et al. examined short congestion cycles while comparing repeated observations in the same elite players [21]. During two-match cycles separated by no more than three days, injury risk was higher during the final 15 min of the second match than during matches outside these cycles. In the final match of three-match congestion cycles, overall injury risk was greater, with additional increases reported for first-half injuries, ankle sprains, and non-contact injuries associated with changes of direction [21].

The systematic review by Page et al. found that match injury incidence generally tended to be higher during congested periods, although findings differed according to the injury outcome, study design, definition of fixture congestion, and competitive setting [13].

4. Discussion

4.1. Main Findings

The evidence included in this narrative review shows that post-match recovery during fixture congestion is not uniform across physiological, neuromuscular, performance, and perceptual domains. The most consistent observation is that the time required to recover depends on both the variable being measured and the competitive context in which recovery takes place. Short inter-match intervals can therefore coincide with incomplete recovery even when selected performance measures have already returned toward pre-match values.

Neuromuscular disturbances were reported for up to 48–72 h in several studies. Nédélec et al. found reductions in CMJ performance, hamstring strength, and peak sprint speed during the post-match recovery period, together with increased muscle soreness [7]. Thomas et al. reported an even slower recovery of selected neuromuscular variables following simulated match-play, with maximal voluntary contraction and peripheral contractile function still not fully restored at 72 h [8]. Competitive match-play produced a somewhat shorter time course in Brownstein et al., where central and peripheral neuromuscular measures recovered within 48 h and maximal voluntary contraction recovered by 72 h, while perceived fatigue remained elevated [9].

Recovery was also affected by the amount and frequency of match exposure. Lundberg and Weckström found no deterioration in sprint or jump performance 72 h after a three-match week, but players with greater match exposure showed larger changes in muscle soreness and plasma urea [3]. In contrast, Hernández-Davó et al. reported impaired CMJ and selected sprint and change-of-

direction measures 72 h after a three-match period compared with a one-match period [5]. Wang et al. further showed that CMJ performance, perceived soreness, and pain sensitivity recovered more slowly during a congested week than during a regular week [11].

These differences are important because fixture congestion cannot be understood simply as a reduction in physical performance. Several studies reported preserved match-running or field-test performance while biochemical or perceptual indicators remained altered [3,10,12,16]. Consequently, the presence or absence of performance impairment alone provides an incomplete description of the recovery process.

4.2. Is 48–72 Hours Sufficient for Complete Recovery?

Whether 48–72 h is sufficient for recovery cannot be answered with a single threshold applicable to all players and all outcome measures. The literature included in this review shows substantial variation between studies, but there is considerable evidence that 48 h may be insufficient for the restoration of several aspects of neuromuscular and perceptual function.

Following competitive match-play, Nédélec et al. reported disturbances in performance and soreness throughout the 72-h observation period [7]. Thomas et al. found persistent neuromuscular impairment beyond 48 h after simulated soccer, with maximal voluntary contraction and peripheral contractile function still below baseline at 72 h [8]. Brownstein et al. reported a faster recovery following an actual competitive match, but maximal voluntary contraction remained reduced at 48 h, and perceptions of fatigue persisted until 72 h [9]. These differences also demonstrate that recovery kinetics may depend on the type of match stimulus and the population investigated.

The findings of Franceschi et al. are particularly relevant to the practical use of the 48-h recovery window. In youth players, CMJ and perceived fatigue had recovered by 48 h, whereas posterior-chain strength and muscle soreness remained impaired [18]. The players were therefore not uniformly recovered at a time when normal training activity could already be resumed. The study also showed that manipulating the training load at MD+2 affected subsequent physical responses, although the authors cautioned that the observed effects were not sufficient to establish that the greater training load critically impaired recovery [18].

Wang et al. reported a similar problem under congested conditions. Compared with a regular week, CMJ height was lower at both 24 and 48 h during the congested week, while pain sensitivity and perceived soreness also showed less favourable recovery profiles [11]. The authors nevertheless

cautioned that incomplete recovery at 48 h should not automatically be interpreted as evidence of poorer performance in a subsequent match.

A 72-h recovery interval was associated with restoration of several variables in some studies, although complete recovery was not consistently observed across all outcome domains. Hernández-Davó et al. observed impaired CMJ, change-of-direction performance, ankle dorsiflexion, and selected sprint measures 72 h after a three-match period [5]. Mohr et al. also found prolonged repeated-sprint impairment and muscle-damage responses during repeated match-play, particularly after the match preceded by the shorter, 72-h recovery interval [14]. Their results indicated that some muscle-damage and inflammatory responses remained altered on the third recovery day following that match [14].

Other studies present a less negative picture. Brownstein et al. found recovery of measured neuromuscular function by 48–72 h after competitive match-play [9], and Lundberg and Weckström observed no impairment in sprint or jump performance 72 h after a congested three-match week [3]. Marqués-Jiménez et al. likewise reported physical-performance values similar to baseline following 72 h of recovery [10]. However, in both the Lundberg and Weckström study and the Marqués-Jiménez study, other indicators remained altered despite the preservation or restoration of physical performance [3,10].

The evidence therefore does not justify treating either 48 or 72 h as a universal point of complete recovery. Forty-eight hours may be insufficient for the recovery of several neuromuscular, muscular, biochemical, and perceptual outcomes, while 72 h may allow restoration of selected performance measures without necessarily normalizing all aspects of recovery.

4.3. Dissociation Between Performance and Physiological Recovery

One of the clearest findings across the included literature is the different time course of recovery between performance-based and non-performance outcomes. The restoration of running, jumping, or sprinting ability did not consistently coincide with normalization of biochemical, hormonal, neuromuscular, or perceptual measures.

Lundberg and Weckström provide a direct example. Sprint and jump performance were not affected 72 h after the congested match cycle, whereas muscle soreness and plasma urea showed greater changes among players with high match exposure [3]. The authors also reported that selected indicators of physiological stress and muscle damage remained evident despite the absence of a measurable reduction in physical fitness.

Marqués-Jiménez et al. observed a comparable separation between physical and hormonal recovery [10]. After 72 h, squat-jump performance and repeated-sprint variables did not differ significantly from baseline. Testosterone and urea were also similar to pre-match values, but cortisol remained elevated, and the testosterone-to-cortisol ratio remained reduced [10]. Physical test performance therefore suggested recovery, while the hormonal response had not returned completely to its pre-match state.

Wilke et al. reported another form of this dissociation during international tournament play [17]. External match-load variables and perceptual responses did not differ significantly between the tournament phases characterized by 48- and 72-h match intervals, yet CK and C-reactive protein responses were greater during the phase with the shorter interval [17]. This difference occurred even though the players were able to maintain several aspects of match performance.

A similar pattern has been reported in female soccer. Goulart et al. found in their systematic review and meta-analysis that physical-performance measures were recovered by 72 h after female soccer matches, whereas CK and lactate dehydrogenase remained elevated at the same time point [22].

The opposite pattern is also possible. Brownstein et al. reported recovery of several objective neuromuscular and performance measures while perceptions of fatigue remained elevated at 72 h [9]. Franceschi et al. found that CMJ and perceived fatigue had returned to baseline after 48 h, while posterior-chain strength and soreness had not [18]. Thus, no single category of marker consistently represents the overall recovery state.

For the interpretation of recovery kinetics, this distinction is central. An athlete may be capable of producing a near-baseline result in a field test or maintaining match-running output while still showing residual disturbances in other domains. Using performance recovery as synonymous with complete recovery would therefore overlook information demonstrated by several of the studies included in this review.

4.4. Cumulative Fatigue During Repeated Match Exposure

The duration of fixture congestion appears to be an important factor in addition to the interval between individual matches. Short periods of congestion do not always result in measurable performance impairment, whereas repeated exposure over several weeks may produce a different response.

Rollo et al. provide the clearest example of this distinction [4]. No deterioration in CMJ, sprint, or Yo-Yo performance was detected after three weeks of playing two matches per week. After six

weeks, however, the two-match group showed poorer jump, sprint, and repeated high-intensity exercise performance, whereas players completing one match per week did not show the same decline [4]. Subjective stress and recovery scores did not change in parallel with these physical outcomes.

The findings suggest that the effects of fixture congestion may become more apparent as repeated exposure continues. Rollo et al. discussed repeated muscular stress as one possible contributor to the cumulative physical-performance decline, while acknowledging that the underlying mechanism was not directly examined in their study [4]. It is therefore more appropriate to describe their findings as evidence of a cumulative performance effect rather than evidence of a specific physiological mechanism.

The three-match sequence investigated by Mohr et al. also demonstrates that a player's response to one match may depend on the matches that preceded it [14]. The slowest recovery of several variables occurred following the second match, despite high-intensity running during that match being slightly lower than in the first and third games. The second match followed a 72-h interval, whereas 96 h separated the second and third matches [14].

Page et al. reproduced a short congested schedule experimentally by exposing players to three soccer-specific simulations separated by 48 h [15]. Reductions in knee-flexor peak torque and increases in muscle soreness were evident across successive trials, indicating residual muscular fatigue during repeated exposure.

At the same time, cumulative fatigue should not be assumed in every congested period. Lundberg and Weckström found no decrement in sprint or jump performance 72 h after a three-match week [3], while Dellal et al. observed stable physical and technical match performance during prolonged periods of successive matches [16]. The response to repeated competition therefore appears to depend on the outcome measured, the duration of exposure, individual match participation, and the recovery time available between matches.

4.5. Role of Match and Training Load

The number of days separating matches is only one element of the recovery process. The studies included in this review also show that individual match exposure and the distribution of training load influence the conditions under which recovery occurs.

Nédélec et al. demonstrated that specific actions performed during competition were related to later recovery responses [7]. Short sprints were associated with greater muscle soreness at 48 and 72 h,

while the number of hard changes of direction was related to the reduction in CMJ performance at 24 h. The authors therefore proposed that conventional measures such as total distance and distance covered at different intensities may not capture all match actions relevant to post-match fatigue.

Individual playing time is also relevant. Garcia et al. divided professional players according to whether they completed at least 60 min, less than 60 min, or did not participate in matches [6].

Players with at least 60 min of match exposure generally accumulated greater workload indices. However, regular weeks showed higher values for several internal and external load indices than congested weeks because training load was reduced when more matches were scheduled [6].

A congested week does not therefore necessarily contain a greater measured training load than a regular week. Oliveira et al. similarly reported that overall weekly sRPE did not differ significantly between one-, two-, and three-match microcycles even though the distribution of training across the week changed substantially [2].

Howle et al. also found lower external training loads during multi-match weeks and lower training loads in seasons containing more congested scheduling [19]. The greater competitive exposure therefore partly replaced normal training activity. Despite this reduction in training load, injury incidence was higher during multi-match weeks [19].

The findings of Franceschi et al. extend this issue into the recovery period itself [18]. At MD+2, posterior-chain function and soreness had not fully recovered. Players assigned to the complete training session accumulated a substantially greater internal and external load than players performing the reduced session, and differences in subsequent physical responses were observed between the two training conditions [18].

Fixture density, match exposure, and training load should therefore be interpreted together. Two players facing the same match schedule may accumulate substantially different competitive loads, while training may be deliberately reduced for one player and increased for another depending on previous match participation.

4.6. Practical Implications for Player Monitoring

The heterogeneity of recovery responses has direct implications for player monitoring. The studies reviewed here do not identify one measure that consistently represents the complete recovery process. Instead, different tools appear sensitive to different components of fatigue and recovery.

Nédélec et al. identified CMJ performance and peak sprint speed as sensitive measures of post-match neuromuscular status among the variables they assessed [7]. They also emphasized the value

of incorporating accelerations, decelerations, and hard changes of direction into match-load analysis because these actions were associated with later recovery responses.

However, CMJ alone may not be sufficient. Franceschi et al. found that CMJ had recovered at 48 h while posterior-chain strength and muscle soreness remained impaired [18]. Brownstein et al. found that objective neuromuscular measures recovered before perceived fatigue [9]. Lundberg and Weckström observed preserved sprint and jump performance despite greater soreness and plasma urea responses after fixture congestion [3]. Wang et al. demonstrated differences between congested and regular weeks using a combination of CMJ, perceived soreness, and pressure pain sensitivity [11].

Taken together, the reviewed studies support the combined interpretation of neuromuscular or physical-performance measures, perceptual responses, and information on individual match and training exposure [3,7,9,11,18]. Biochemical measures may provide additional information where they are available, but the studies also show that their recovery profile may differ substantially from field-based performance.

The interpretation of monitoring data should also remain individual. Nédélec et al. noted that players exposed to the same team environment may show different recovery responses and highlighted potential interindividual differences in recovery capacity [7]. Lundberg and Weckström likewise observed that not every highly exposed player showed the same biochemical or soreness response following congestion [3]. Individual longitudinal changes may therefore be more informative than assuming that every player should have recovered after a predetermined number of hours.

4.7. Implications for Injury Prevention

The association between fixture congestion and injury has important implications for player management, but it should not be interpreted as proof that incomplete recovery directly causes injury. The studies included in this review were mainly observational with respect to injury outcomes, and the magnitude of risk differed across investigations.

Dupont et al. reported a substantial increase in injury incidence when professional players completed two matches per week compared with one match per week, despite no significant deterioration in the match-running variables assessed [12]. Bengtsson et al. found higher total and muscle injury rates in league matches preceded by four or fewer recovery days than in matches with at least six recovery days, with hamstring and quadriceps injuries particularly affected [20].

Carling et al. showed that injury risk could also vary within the match itself during short congestion cycles [21]. In two-match cycles, injury risk was higher during the final 15 min of the second match. In three-match cycles, overall injury risk was greater in the final match, together with increased risks for selected injury patterns, including ankle sprains and non-contact injuries associated with changes of direction [21]. The authors identified player substitution and rotation as potential practical considerations during short periods of match congestion.

Howle et al. reported higher injury incidence during multi-match weeks even though training loads were reduced [19]. Dellal et al., however, did not find a significant difference in overall injury incidence between congested and non-congested periods; the pattern differed when match and training injuries were analyzed separately, with match injury incidence higher and training injury incidence lower during congestion [16].

This inconsistency is also reflected in the systematic review by Page et al. [13]. Although the evidence generally indicated greater match injury risk during congested periods, results varied according to the definition of congestion, injury outcome, competition, and study design.

For injury prevention, the available evidence therefore supports attention to match exposure and recovery status during congested periods, but it does not justify using a single recovery marker as a direct injury predictor. The persistence of soreness, strength deficits, or other signs of incomplete recovery may provide useful contextual information, while player rotation and individual exposure management may help reduce repeated demands on the same players. These measures should be viewed as components of player management rather than as proven methods for eliminating congestion-related injury risk.

4.8. Population-Specific Considerations

Most of the direct fixture-congestion evidence included in this review was obtained from male players. Several studies involved professional senior players [2–3,6,12,16,19–21], while others examined semi-professional or subelite populations [4,8–9,15]. The results cannot therefore be assumed to apply equally across all competitive levels.

Youth players were represented in several recent studies. Hernández-Davó et al. reported reduced physical performance 72 h after a three-match period in young players [5]. Wilke et al. examined the Brazilian Under-20 national team during an international tournament and found greater biochemical disturbances during the phase with 48-h match intervals than during the phase with 72-h intervals [17]. Franceschi et al. showed that posterior-chain strength and soreness remained

impaired at 48 h in Italian Serie A youth players [18], while Wang et al. reported slower CMJ and pain-related recovery during a congested week in under-19 players [11]. These findings indicate that incomplete recovery during short match intervals is also relevant in younger populations, although the specific response is not identical across studies.

Female players remain less represented within the fixture-congestion literature used for the present review. The female-specific publication in the predefined bibliography is the systematic review and meta-analysis by Goulart et al. [22], which focused on post-match recovery rather than fixture congestion itself. The review included 26 studies and 465 female players. Physical-performance measures were generally recovered by 72 h, while CK and lactate dehydrogenase remained increased at this time point. The authors also emphasized the limited amount of evidence describing extended recovery periods in female soccer, particularly for perceptual outcomes [22].

The female findings are consistent with the broader theme identified in the present review—that recovery of physical performance may occur before complete normalization of other markers—but direct evidence on repeated congested match exposure in female players remains limited within the bibliography examined here. Extrapolation of findings derived primarily from male professional soccer to women's soccer should therefore be made cautiously.

Differences between youth, subelite, and professional populations should also be considered when interpreting recovery time courses. Training history, competitive demands, match exposure, and the type of recovery assessments differed considerably between the included studies. Consequently, the available evidence supports interpreting recovery in relation to the population and competitive setting in which the measurements were obtained rather than applying a single recovery timeline to all soccer players.

4.9. Limitations of Current Evidence

The current evidence on fixture congestion and post-match recovery in soccer has several limitations that should be considered when interpreting the findings of this review. Direct comparison between studies is complicated by substantial differences in study populations, definitions of fixture congestion, competitive settings, recovery time points, and outcome measures.

One important limitation is the relatively small number of players included in many original investigations. Several studies assessing recovery kinetics were conducted with small samples drawn from a single team or competitive environment. Nédélec et al., for example, examined ten

professional outfield players across four competitive matches [7], while Thomas et al. studied 15 semi-professional players using a simulated match protocol [8]. Brownstein et al. included 16 semi-professional players [9], and Lundberg and Weckström assessed 16 professional players after congested and regular match cycles [3]. These small samples limit the extent to which the observed recovery profiles can be generalized to other teams, competitive levels, and playing populations. Nédélec et al. also noted that recovery potential may differ between players exposed to the same team environment [7].

The definition and practical representation of fixture congestion also varied considerably. Julian et al. defined congestion as at least two successive matches separated by less than 96 h [1], whereas individual studies used schedules ranging from two matches per week to three matches within a week, repeated matches separated by 48 h, comparisons between 48- and 72-h intervals, and longer periods involving repeated congested weeks [3–6,11,14–17,19]. Studies grouped under the general concept of fixture congestion therefore did not necessarily expose players to comparable competitive schedules. This problem is also evident in the systematic review by Julian et al., where only a limited number of studies used sufficiently comparable variables for quantitative synthesis and heterogeneity remained an important methodological issue [1].

Differences in the timing of recovery assessments create another difficulty. Some studies measured players immediately after match-play and then at 24, 48, and 72 h [7–9], whereas others assessed only one selected recovery point or compared responses between different competitive schedules. Lundberg and Weckström focused on measurements obtained 72 h after the final match of congested and regular cycles [3], while Wang et al. followed recovery only until 48 h [11]. Wilke et al. compared tournament phases characterized by 48- and 72-h inter-match intervals but collected biochemical samples approximately 13–15 h after the individual matches [17]. These designs address related aspects of recovery but are not methodologically equivalent. A measurement obtained 48 h after a match cannot be interpreted in the same way as a measurement collected approximately 13–15 h after a match occurring within a 48-h competitive schedule.

The recovery outcomes themselves were highly heterogeneous. The included studies used CMJ and squat jump, sprint tests of different distances, repeated-sprint protocols, maximal voluntary contraction, isolated muscle-strength assessments, creatine kinase, urea, inflammatory markers, hormonal variables, muscle soreness, perceived fatigue, recovery questionnaires, pressure pain sensitivity, and internal and external workload measures. Even within a single category such as

jump performance, different testing variables may show different sensitivity to fatigue. Franceschi et al., for example, reported that CMJ-derived variables differed in their response to match-related fatigue [18]. This methodological diversity makes direct comparison of recovery timelines more difficult.

The inclusion of both simulated and competitive match-play introduces an additional source of variability. Laboratory simulations provide greater experimental control but do not reproduce every demand of competitive soccer. Brownstein et al. noted that simulated protocols do not fully replicate the perceptual demands of decision-making, reacting and anticipating, or the complete range of mechanically and neuromuscularly demanding actions occurring during real match-play [9]. This distinction is relevant when comparing their competitive-match findings with the slower recovery reported by Thomas et al. following simulated match-play [8–9]. Brownstein et al. observed a substantially faster recovery of contractile function after competitive soccer and discussed differences in the match protocol and phase of the season as possible explanations for the discrepancy [9].

Competitive matches, however, provide less experimental control over individual match exposure. Brownstein et al. emphasized that competitive match-play is inherently unpredictable and that substantial inter-individual variability exists in match activity, which can influence both the magnitude of fatigue and the subsequent recovery time course [9]. Nédélec et al. similarly identified relationships between specific match actions and later recovery responses but acknowledged that their video analysis quantified the frequency rather than the intensity of the actions performed [7].

Match type represents a related limitation. The literature includes official competitive matches, tournament settings, simulated matches, and friendly match-play. Franceschi et al. used a friendly match in youth players and acknowledged that, although this increased ecological validity compared with a laboratory protocol, it reduced experimental control over the match stimulus [18]. League, cup, tournament, and friendly matches represent different competitive settings, while differences in match type, competition level, scheduling, and club practices may contribute to variability between studies [13]. Page et al. identified these methodological and contextual differences as possible explanations for inconsistent findings across the fixture-congestion literature [13].

Individual match exposure further complicates interpretation. Fixture congestion describes a team schedule, whereas the physiological load experienced by individual players depends partly on their actual participation. Garcia et al. demonstrated differences in workload according to match participation [6], while Lundberg and Weckström distinguished between highly exposed players and non-selected controls [3]. Studies that do not account adequately for minutes played may therefore include players with substantially different competitive exposure within the same congested schedule. The systematic review by Page et al. also identified inconsistent treatment of individual playing duration as an important methodological limitation in the injury literature [13]. Squad rotation may further influence the observed effects of congestion. Teams with greater squad depth may distribute competitive exposure across more players, whereas other teams may rely more heavily on the same individuals. Howle et al. noted that differences in sample size and squad depth, including the possibility of greater player rotation in some professional clubs, may partly explain differences in injury findings between studies [19]. The number of matches scheduled within a week therefore does not necessarily correspond to the same individual exposure for all players.

Training performed between matches is another factor that is not standardized across the available studies. During congested periods, training load may be reduced or redistributed according to the number of matches and individual match participation [2,6,19]. Franceschi et al. directly examined this issue and showed that training-load manipulation at 48 h post-match influenced subsequent physical responses in youth players [18]. Differences in training performed during the recovery period may therefore contribute to variability in recovery profiles between studies.

Recovery-related behaviours and interventions are also difficult to control in field settings. Nutrition was not recorded throughout the six-week observational period in Rollo et al. [4], whereas Nédélec et al. attempted to standardize nutritional practices and asked participants not to use additional recovery treatments during the post-match period [7]. In Wang et al., sleep, nutrition, and hydration remained part of the players' normal unsupervised routines, and structured recovery strategies were not implemented during the observation period [11]. Wilke et al. also acknowledged that recovery can be influenced by factors that were not fully assessed in their study, including sleep, nutrition, mood, aerobic capacity, and strength [17]. Such factors make it difficult to attribute all differences in recovery exclusively to fixture density.

The available evidence is also dominated by male players. Although the bibliography includes studies of youth populations and the systematic review and meta-analysis of female soccer by

Goulart et al. [22], direct evidence comparing congested and regular schedules in female players remains limited. Goulart et al. noted that the number of available studies on extended post-match recovery in women was relatively small, preventing subgroup analyses according to playing level or single-match versus congested-schedule conditions. The number of available observations also decreased at later recovery time points [22]. Findings derived primarily from male professional or semi-professional players should therefore be generalized to women's soccer with caution.

Injury-related evidence has additional limitations. Higher injury incidence has been reported in several congested settings [12,19–21], but these associations do not establish that residual fatigue directly caused the injuries. Howle et al. explicitly stated that it remained unclear whether increased injury rates were related to reduced recovery, greater match exposure, or the specific external demands of the matches [19]. Page et al. also reported considerable heterogeneity in injury-surveillance methods, individual exposure, team characteristics, competitive schedules, and reporting procedures [13]. Injury findings should therefore be interpreted as associations with congested scheduling rather than evidence of a direct causal pathway from incomplete recovery to injury.

Overall, the available literature indicates that post-match recovery is heterogeneous and may vary considerably between individuals. Differences in the definition of fixture congestion, study design, player characteristics, match exposure, training load, recovery practices, measurement methods, and assessment timing all contribute to variation in the reported recovery kinetics. These limitations should be considered when interpreting fixed recovery periods such as 48 or 72 h and when applying findings obtained in one competitive environment to another.

4.10. Future Research Directions

Future research on fixture congestion and post-match recovery should aim to improve methodological consistency and provide a clearer description of the recovery time course across different competitive settings. One of the main priorities is greater standardization of how fixture congestion is defined. The studies included in the present review used different criteria, including two or more matches within a week, three-match weeks, inter-match intervals of 48 or 72 h, and definitions based on successive matches separated by less than 96 h. Julian et al. also identified considerable methodological heterogeneity across studies examining fixture congestion [1]. Clear reporting of the number of matches, exact inter-match intervals, individual playing time, and the duration of the congested period would facilitate more accurate comparison between studies.

Direct comparisons of specific recovery intervals remain particularly important. Comparisons of 48-, 72-, and 96-h recovery periods within comparable competitive or experimental conditions could help clarify how the duration of the interval relates to subsequent recovery. Wilke et al. provided useful data from tournament phases characterized by 48- and 72-h inter-match intervals [17], but their study was observational and conducted within a single youth national-team tournament. The authors acknowledged the limited capacity for extrapolation and the influence of factors that were not fully controlled [17]. Repeated observations of the same players under different recovery intervals may help reduce some of the variability between separate study populations.

Larger study populations are also required. Several investigations of recovery kinetics involved relatively small groups from individual teams [3,7–9,11]. Wang et al. specifically acknowledged that their relatively small sample limited the generalizability of the findings to other competitive settings [11]. Dellal et al. likewise indicated that a larger sample, together with consideration of contextual factors affecting match performance, could improve the detection of changes during periods of fixture congestion [16]. Studies involving larger samples and repeated observations across different competitive periods would therefore strengthen the generalizability of the available evidence.

More evidence is particularly needed in female soccer. Most direct investigations of fixture congestion in the present literature involve male players. Goulart et al. reported that the number of studies examining extended post-match recovery in female players remained limited, and the available evidence did not allow subgroup analyses according to playing level or single-match versus congested-schedule conditions [22]. Wang et al. also identified the exclusive inclusion of male players as a limitation and indicated the need to investigate congested schedules in female athletes [11]. Direct research on post-match recovery during fixture congestion in female players is therefore needed before recovery timelines derived primarily from male populations can be applied more broadly.

Further work is also required to compare youth and senior players. The available youth studies show that selected recovery measures may remain altered within the first 48–72 h after match-play [5,11,17–18]. However, differences in populations, testing procedures, competitive environments, and recovery measures make comparison with senior players difficult. Page et al. also highlighted the need for additional research involving both female and youth players exposed to congested

schedules [13]. Direct comparisons between youth and senior players are needed to clarify whether recovery responses differ between these populations [5,13,17–18].

Another important direction concerns multidimensional assessment of recovery. The evidence synthesized in this review indicates that physical performance, neuromuscular function, biochemical responses, and perceptual recovery do not necessarily return to baseline at the same time. A single outcome may therefore provide only a partial description of recovery status. Dellal et al. proposed combining objective fatigue markers with subjective measures to provide additional information on individual and team responses to fixture congestion [16]. Wang et al. similarly noted that the inclusion of additional objective physiological measures could provide a more comprehensive assessment of recovery [11]. Research incorporating physical or neuromuscular tests together with perceptual and, where available, physiological or biochemical measures would therefore allow a broader description of post-match recovery.

Individual recovery trajectories also warrant greater attention. Nédélec et al. emphasized that players within the same team may differ in their recovery potential and proposed repeated observations of the same players across several matches to investigate intra-individual differences in recovery kinetics [7]. Longitudinal monitoring of individual players could help determine whether recovery responses remain consistent across matches or vary according to match exposure and competitive circumstances. Individual playing time, match actions, and training performed during the recovery period should be considered alongside recovery outcomes.

Longer observation periods are another important area. Much of the available literature examines isolated congested weeks or relatively short sequences of matches. Rollo et al. found no impairment of the assessed physical-performance variables after three weeks of two matches per week, whereas significant reductions in jump, sprint, and repeated high-intensity exercise performance were observed after six weeks [4]. Dellal et al. also called for further investigation of physiological responses during prolonged periods of fixture congestion and across different stages of the season [16]. Longer-term observations are therefore required to determine how recovery and performance responses change when short inter-match intervals are repeatedly imposed.

The interaction between post-match recovery and training performed between fixtures also deserves further investigation. Franceschi et al. found that selected recovery measures had not returned to baseline at 48 h and recommended additional research examining recovery- and training-based strategies during the first two days after match-play, including repeated-measures

and crossover approaches [18]. More detailed monitoring of training load together with repeated recovery assessments could help clarify how training performed during MD+1 and MD+2 relates to subsequent recovery.

Finally, the relationship between residual fatigue and later injury requires more direct investigation. Current epidemiological studies demonstrate associations between fixture congestion and injury occurrence, but they do not establish that incomplete recovery is the mechanism responsible for increased injury risk [13,19–21]. Page et al. highlighted inconsistencies in injury-surveillance methods and the need to account more carefully for playing duration, position, and other contextual factors [13]. Combining prospective injury surveillance with repeated measurements of neuromuscular, perceptual, and physiological recovery could help clarify whether persistent disturbances are associated with injuries occurring during subsequent match or training exposure. Until such evidence is available, residual fatigue and injury risk should be treated as related but distinct outcomes rather than as a confirmed causal sequence.

Overall, future work should focus on more consistent definitions of fixture congestion, larger and more diverse study populations, direct comparisons of specific recovery intervals, repeated observations within the same players, and simultaneous assessment of objective and subjective recovery measures. Longer-term monitoring across repeated congested periods may help determine how recovery and performance responses change when short inter-match intervals are repeatedly imposed.

4.11. Practical Applications

Recovery during fixture congestion should not be assessed using a single performance variable. Studies included in this review demonstrate that physical performance may return toward pre-match values while other indicators remain altered. Lundberg and Weckström found no clear impairment in sprint or jump performance 72 h after a congested three-match week, despite greater changes in muscle soreness and plasma urea among highly exposed players [3]. Marqués-Jiménez et al. similarly reported physical performance comparable with baseline after 72 h, while cortisol remained elevated and the testosterone-to-cortisol ratio remained reduced [10]. Restoration of performance in an isolated test should therefore be interpreted together with other indicators of recovery.

Monitoring can incorporate information from different recovery domains. Nédélec et al. showed that CMJ performance, hamstring strength, peak sprint speed, and muscle soreness remained

affected during the 72-h post-match period [7]. They also found that short sprints and hard changes of direction performed during the match were associated with subsequent soreness and CMJ impairment [7]. Their findings indicate that total distance and running distance at different intensities do not fully describe the match actions associated with subsequent recovery. Accelerations, decelerations, and demanding changes of direction should also be considered when individual match exposure is evaluated [7].

A practical interpretation of recovery may therefore combine neuromuscular or muscular measures, such as CMJ or posterior-chain strength, with perceptual responses such as soreness or fatigue and information on individual match and training exposure [6–7,11,18]. The rationale for combining these measures is supported by the different recovery patterns observed across the included studies rather than by the assumption that one marker reflects the player's complete recovery status.

Individual match participation is particularly important when training load is planned during congested weeks. Garcia et al. classified professional players according to whether they completed at least 60 min, less than 60 min, or did not participate in matches and found clear differences in accumulated workload between these groups [6]. Players completing at least 60 min generally showed the highest values for most workload indices [6]. The authors emphasized the importance of monitoring internal and external load according to match participation and reducing disparities between players exposed to different competitive loads [6]. Training-load management should consequently account for differences in individual match participation, particularly between players with high match exposure and those completing few or no match minutes [6].

The second day after match-play appears to be an important point for recovery monitoring and training-load management. Franceschi et al. assessed youth players before and after match-play and again at 48 and 72 h, with different training-load interventions applied at MD+2 [18]. At 48 h, CMJ and perceived fatigue had returned to baseline, whereas isometric posterior-chain function and muscle soreness had not fully recovered [18]. These findings show that the normalization of selected variables at MD+2 does not necessarily coincide with recovery of all measured outcomes. Decisions regarding training load at this point should therefore consider more than one recovery measure.

Wang et al. provide additional evidence for monitoring during the first 48 h of congested schedules [11]. In their study, the congested week consisted of two matches within seven days separated by no more than three days, and players were assessed before and immediately after the match and

again at 24 and 48 h [11]. Recovery of CMJ performance and pain-related measures was less favourable during the congested week than during the regular week. Combining functional and perceptual assessments may therefore provide a broader view of short-term recovery than relying on either type of measure alone [11].

The interpretation of recovery at 72 h also requires caution. Hernández-Davó et al. observed impairments in jumping, selected sprint measures, change-of-direction performance, and ankle dorsiflexion range of motion 72 h after a three-match period compared with a one-match period [5]. The authors concluded that players were not completely recovered at this time following the congested schedule and highlighted the value of monitoring during such periods [5]. Other studies, however, reported restoration of selected physical-performance variables by 72 h [3,9–10]. A 72-h interval should therefore not be considered a universal indicator of complete recovery; its interpretation depends on the variables assessed and the preceding match exposure.

Repeated congestion over several weeks should also be considered. Rollo et al. found no significant deterioration in the assessed physical-performance measures after three weeks of playing two matches per week. After six weeks, however, the two-match group showed poorer CMJ, sprint, and repeated high-intensity exercise performance [4]. Short-term maintenance of performance during one congested period does not necessarily describe the response to repeated exposure over a longer period. Monitoring may be particularly relevant for players who continue to accumulate substantial match minutes across successive congested weeks.

Match exposure should also be considered when managing players in relation to injury risk. Howle et al. reported higher injury incidence during multi-match weeks despite reductions in external training load [19]. Their practical applications included consideration of player rotation during periods of increased fixture congestion and quantification of match loads to improve understanding of individual exposure [19]. These findings do not demonstrate that incomplete recovery directly causes injury, but they support careful management of repeated match exposure during congested schedules.

In practice, recovery assessment during fixture congestion should integrate individual match exposure, training load, neuromuscular or muscular performance, and perceptual responses [6–7,11,18]. Particular attention should be paid to recovery status around MD+2 and to players repeatedly exposed to high match loads. Restoration of a single performance variable, or the

completion of a fixed 72-h recovery period, should not be interpreted in isolation as evidence of complete recovery [3,5,10].

5. Conclusions

Fixture congestion can modify post-match recovery in soccer players, particularly when short recovery periods are combined with repeated match exposure [3–5,11]. The available evidence indicates that recovery is a multidimensional and heterogeneous process, with neuromuscular, physical, biochemical, and perceptual responses following different time courses [3,7–11,18].

A recovery period of 48 h may be insufficient for the restoration of selected measures of muscle function, neuromuscular performance, soreness, and perceived recovery [7–9,11,18]. At 72 h, several performance variables had returned toward baseline in some studies and competitive settings, but complete recovery was not consistently observed across all outcome domains [3,5,8–10]. Lundberg and Weckström reported preserved sprint and jump performance despite residual muscle soreness and physiological stress [3], while Marqués-Jiménez et al. found physical performance similar to baseline after 72 h despite persistent alterations in cortisol and the testosterone-to-cortisol ratio [10].

Therefore, a return of physical performance does not necessarily indicate complete recovery. Fixed recovery periods such as 48 or 72 h should not be interpreted in isolation as evidence of complete recovery. Although increased injury occurrence has been reported during some congested schedules, the available evidence does not establish incomplete recovery as its direct cause [13,19–21]. Recovery assessment during congested schedules should therefore integrate information on individual match exposure, workload, physical or neuromuscular performance, and perceptual responses rather than relying on a single marker or a predetermined recovery interval [6–7,11,18].

Disclosure

Supplementary Materials

Not applicable.

Author Contributions

Conceptualization, K.L.; methodology, K.L.; formal analysis, K.L., J.G., J.D. and D.G.; investigation, J.D., A.K. and Ł.I.; resources, D.G., A.K. and Ł.I.; visualization, A.K.; writing—

original draft, J.G., A.K.L. and L.I.; writing—review and editing, K.L.; supervision, K.L.; project administration, J.G. All authors have read and agreed to the published version of the manuscript.

Funding

The study received no external funding.

Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

Not applicable.

Acknowledgements

None.

Conflicts of Interest

The authors declare no conflict of interest in relation to this study.

AI Statement

Artificial intelligence tools have not been used to produce or alter the scientific content of the paper. Publicly available AI-powered websites have been used to identify better wording for certain expressions and sentences to ensure a smooth reading experience, while having no impact on the interpretation of cited evidence.

References:

1. Julian R, Page RM, Harper LD. The effect of fixture congestion on performance during professional male soccer match-play: a systematic critical review with meta-analysis. *Sports Med.* 2021;51:255-273. <https://doi.org/10.1007/s40279-020-01359-9>
2. Oliveira R, Brito J, Martins A, Mendes B, Calvete F, Carriço S, et al. In-season training load quantification of one-, two- and three-game week schedules in a top European professional soccer team. *Physiol Behav.* 2019;201:146-156. <https://doi.org/10.1016/j.physbeh.2018.11.036>
3. Lundberg TR, Weckström K. Fixture congestion modulates post-match recovery kinetics in professional soccer players. *Res Sports Med.* 2017;25:408-420. <https://doi.org/10.1080/15438627.2017.1365296>

4. Rollo I, Impellizzeri FM, Zago M, Iaia FM. Effects of 1 versus 2 games a week on physical and subjective scores of subelite soccer players. *Int J Sports Physiol Perform*. 2014;9(3):425-431. <https://doi.org/10.1123/IJSP.2013-0288>
5. Hernández-Davó JL, Moreno-Pérez V, Moreno Navarro P. Effects of playing 1 vs 3 matches in a one-week period on physical performance in young soccer players. *Biol Sport*. 2022;39(4):819-823. <https://doi.org/10.5114/biolSport.2022.108700>
6. Garcia GR, Gonçalves LGC, Clemente FM, Nakamura FY, Nobari H, Bedo BLS, et al. Effects of congested fixture and matches' participation on internal and external workload indices in professional soccer players. *Sci Rep*. 2022;12:1864. <https://doi.org/10.1038/s41598-022-05792-w>
7. Nédélec M, McCall A, Carling C, Le Gall F, Berthoin S, Dupont G. The influence of soccer playing actions on the recovery kinetics after a soccer match. *J Strength Cond Res*. 2014;28(6):1517-1523. <https://doi.org/10.1519/JSC.0000000000000293>
8. Thomas K, Dent J, Howatson G, Goodall S. Etiology and recovery of neuromuscular fatigue following simulated soccer match-play. *Med Sci Sports Exerc*. 2017;49(5):955-964. <https://doi.org/10.1249/MSS.0000000000001196>
9. Brownstein CG, Dent JP, Parker P, Hicks KM, Howatson G, Goodall S, Thomas K. Etiology and recovery of neuromuscular fatigue following competitive soccer match-play. *Front Physiol*. 2017;8:831. <https://doi.org/10.3389/fphys.2017.00831>
10. Marqués-Jiménez D, Calleja-González J, Arratibel-Imaz I, Terrados N. Biochemical and physical performance responses to a soccer match after a 72-hour recovery period. *Sports*. 2022;10:140. <https://doi.org/10.3390/sports10100140>
11. Wang X, Zhao Y, Duan Y, Qiu Q. Recovery kinetics following matches in congested vs. non-congested schedules: effects on pressure pain sensitivity, delayed onset muscle soreness, and muscular performance in soccer players. *BMC Sports Sci Med Rehabil*. 2026. <https://doi.org/10.1186/s13102-026-01813-7>
12. Dupont G, Nedelec M, McCall A, McCormack D, Berthoin S, Wisløff U. Effect of 2 soccer matches in a week on physical performance and injury rate. *Am J Sports Med*. 2010;38(9):1752-1758. <https://doi.org/10.1177/0363546510361236>

13. Page RM, Field A, Langley B, Harper LD, Julian R. The effects of fixture congestion on injury in professional male soccer: a systematic review. *Sports Med.* 2023;53:667-685. <https://doi.org/10.1007/s40279-022-01799-5>
14. Mohr M, Draganidis D, Chatzinikolaou A, Barbero-Álvarez JC, Castagna C, Douroudos I, et al. Muscle damage, inflammatory, immune and performance responses to three football games in 1 week in competitive male players. *Eur J Appl Physiol.* 2016;116(1):179-193. <https://doi.org/10.1007/s00421-015-3245-2>
15. Page RM, Marrin K, Brogden CM, Greig M. Physical response to a simulated period of soccer-specific fixture congestion. *J Strength Cond Res.* 2019;33(4):1075-1085. <https://doi.org/10.1519/JSC.0000000000002257>
16. Dellal A, Lago-Peñas C, Rey E, Chamari K, Orhant E. The effects of a congested fixture period on physical performance, technical activity and injury rate during matches in a professional soccer team. *Br J Sports Med.* 2015;49(6):390-394. <https://doi.org/10.1136/bjsports-2012-091290>
17. Wilke CF, Coimbra CC, Drummond FR, Drummond LR, Campos HO, Kanope T, Ramos GP. Differences between 48 and 72-hour intervals on match load and subsequent recovery: a report from the Brazilian under-20 national football team. *Front Sports Act Living.* 2024;5:1164454. <https://doi.org/10.3389/fspor.2023.1164454>
18. Franceschi A, Robinson MA, Owens DJ, Brownlee T, Connolly DR, Coutts AJ, et al. Post-match recovery responses in Italian Serie A youth soccer players: effects of manipulating training load 48 h after match play. *Eur J Sport Sci.* 2025;25:e12297. <https://doi.org/10.1002/ejsc.12297>
19. Howle K, Waterson A, Duffield R. Injury incidence and workloads during congested schedules in football. *Int J Sports Med.* 2020;41(2):75-81. <https://doi.org/10.1055/a-1028-7600>
20. Bengtsson H, Ekstrand J, Häggglund M. Muscle injury rates in professional football increase with fixture congestion: an 11-year follow-up of the UEFA Champions League injury study. *Br J Sports Med.* 2013;47(12):743-747. <https://doi.org/10.1136/bjsports-2013-092383>
21. Carling C, McCall A, Le Gall F, Dupont G. The impact of short periods of match congestion on injury risk and patterns in an elite football club. *Br J Sports Med.* 2016;50(12):764-768. <https://doi.org/10.1136/bjsports-2015-095501>

22. Goulart KNO, Coimbra CC, Campos HO, Drummond LR, Ogando PHM, Brown G, et al. Fatigue and recovery time course after female soccer matches: a systematic review and meta-analysis. Sports Med Open. 2022;8:72. <https://doi.org/10.1186/s40798-022-00466-3>